

# KINJAL SINGH

## Computer Science & Statistics at UIUC

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**EDUCATION:** University of Illinois at Urbana-Champaign (UIUC)  
*Bachelor of Science in Computer Science and Statistics*

Expected May 2026

### TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript, SQL, Rust, Bash

**Backend & Distributed Systems:** gRPC, REST APIs, Kafka, ELK Stack, Redis Streams, MongoDB, Hadoop

**Database:** Apache Spark, PySpark, Airflow, Kafka, AWS Glue, Kinesis, Google Pub/Sub, Redshift, Snowflake, BigQuery, Athena, PostgreSQL, DynamoDB

**ML & AI:** PyTorch, ONNX, TensorFlow, JAX, Hugging Face, Whisper, LangChain, XGBoost, Scikit-learn, MLflow

**Cloud & DevOps:** AWS (Lambda, EC2, S3, DynamoDB, EKS), GCP, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Istio, Linux

### WORK EXPERIENCE

**Software Engineer Intern - PricewaterhouseCoopers (PwC)**

June 2025 – August 2025

- Built a self-healing **Agentic AI Ops platform** that analyzed real-time logs (**Kafka, ELK, Prometheus**) with **LSTM & Isolation Forest** models to detect and auto-remediate infra incidents.
- Deployed **predictive scaling** using **ARIMA & Prophet** integrated with **Kubernetes HPA** and **Terraform**, reducing downtime by **40%** and manual remediation by **60%**.
- Implemented **C++ gRPC microservices** for LLM inference, optimizing **ONNX runtime** and cutting latency **4×** vs Python-only deployments.

**NLP Research Assistant- Respeaker Project at UIUC**

February 2024 – Present

- Built an edge ML pipeline with **Raspberry Pi** and **ReSpeaker** to process **40+ hrs** of classroom audio, enabling real-time speaker diarization & engagement analytics.
- Improved **sentiment model F1 score from 65% to 78%** using fine-tuned **DistilBERT & Wav2Vec2** on Hugging Face with **PyTorch Lightning** + **MLflow** for reproducible training.
- Optimized audio preprocessing with **Apache Spark** and **Airflow**, cutting latency **35%**, and accelerated large-scale queries to **<200ms** using **BigQuery** and **Athena**.

**Data Engineering Intern - Clover Infotech**

May 2024 – July 2024

- Engineered distributed ETL workflows** with **PySpark**, **AWS Glue**, consolidating **5+ enterprise data sources** into **Snowflake** and **Redshift** for real-time analytics.
- Developed **JavaScript-based Lambda functions** for dynamic schema evolution and data validation, **reducing ETL failures by 25%**.
- Built a **Python Flask service** exposing ETL metadata as **REST APIs** for BI dashboards, enabling **~1M daily API requests**.
- Improved pipeline observability with **Prometheus** and **Grafana** and **Linux-based log rotation strategies**, reducing incident response time by **40%**.

### PROJECT HIGHLIGHTS

**AutoLitigator – Agentic AI-Powered Legal API Platform**

January 2025 - March 2025

- Designed and deployed a **retrieval-augmented(RAG) backend** (**Pinecone** and **Redis**) handling **10K+ documents** with **sub-500ms query latency**.
- Built scalable microservices (**LangChain** and **FastAPI**) containerized on **Kubernetes (EKS)** with **HPA autoscaling**, sustaining **1K+ concurrent requests at 95% uptime**.
- Integrated **CourtListener** + **SEC EDGAR APIs** for automated case triage, achieving **86% relevance accuracy**.

**PolyCast: Real-Time AI Debate Coach**

April 2025 – June 2025

- Built a low-latency streaming architecture using **Whisper ASR**, **Spark Structured Streaming**, and **Rust microservices** to process **10K+ audio chunks <200ms delay**.
- Developed a **multi-service backend** with **C++ gRPC** and **Redis Streams** for scalable debate session analytics.
- Orchestrated **Kubernetes** deployments with **Istio service mesh** for fine-grained routing & load balancing, debugging network bottlenecks at the **Linux container level (iptables, cgroups)**.

**CERTIFICATIONS:** AWS Machine Learning Specialty | AWS Data Engineer Associate | AWS AI Practitioner | AWS Cloud Practitioner | Google Cloud Data Engineer | Google Cloud Digital Leader | Google Cloud Gen AI Leader

